VISHESH KUMAR

Full Stack Developer | Android & Flutter Developer

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Portfolio Website

PROFESSIONAL SUMMARY

BCA undergraduate and practical Software Developer with hands-on experience architecting production web applications, cross-platform mobile systems, and native Android apps. Proficient in Next.js, Flutter, Kotlin (MVVM), and PostgreSQL with Supabase Row Level Security. Proven ability to design secure relational schemas, eliminate race conditions in concurrent reward wallets, and integrate third-party APIs. Dedicated to clean architecture, automated workflows, and continuous engineering improvement.

TECHNICAL SKILLS

Languages:	Dart, Kotlin, TypeScript, JavaScript, Python, Java, C/C++, SQL
Mobile Development:	Flutter, Android SDK, Jetpack Compose, XML Layouts, MVVM Architecture, Room Database, Coroutines, StateFlow
Web & Frontend:	Next.js 14 (App Router), React.js, Tailwind CSS, HTML5, CSS3, Responsive Design, State Management
Backend & Databases:	PostgreSQL, Supabase (RLS, Edge Functions, Auth), Node.js, RESTful APIs, Firebase, SQLite
Developer Tools:	Git, GitHub, VS Code, Android Studio, Postman, Docker, Cloudflare Workers, Vercel, Linux / Bash
AI Engineering Suite:	Claude Code, Google Antigravity, Cursor, GitHub Copilot (with Strict Human Code Ownership & Audit)

TECHNICAL PROJECTS

Earn Master – Mobile Gaming & Rewards Platform — **Flutter, Supabase, PostgreSQL | Live: earn-master.online**

- Architected and deployed a production cross-platform mobile gaming app featuring wallet ledger, reward redemption, and task tracking.
- Implemented atomic database transactions and PostgreSQL advisory locks with Row Level Security (RLS) to prevent race conditions and credit duplication.
- Engineered real-time state synchronization and connected Supabase Edge Functions for secure third-party reward callback processing.

Restaurant Lead Finder – Geospatial Discovery Platform — **Next.js 14, Mapbox GL, Supabase, Overpass API**

- Built full-stack B2B lead discovery platform with interactive map clustering, multi-criteria filtering, and deduplication logic.
- Implemented bounding-box spatial queries and client-side viewport debouncing, reducing unnecessary API re-queries by 40%.
- Developed automated data extraction and CSV/JSON export workflows for structured sales pipeline management.

Code Master – Native Android Programming Learning App — **Kotlin, Jetpack Compose, MVVM, Room DB, Coroutines**

- Engineered offline-first native Android educational application providing programming tutorials, code snippets, and interactive quizzes.
- Structured codebase using clean MVVM architecture, Kotlin Coroutines, and StateFlow for fluid, non-blocking UI rendering.
- Designed local persistence schema using Android Room DB to support zero-latency offline learning and progress persistence.

Online Code Compiler & Sandboxed Execution IDE — **React, Node.js, Express, Docker Sandbox, Judge0 API**

- Developed a web-based multi-language code compilation environment supporting Python, Java, C++, and JavaScript.
- Implemented containerized execution isolation with strict 5-second process timeouts and memory ceilings to guard against malicious fork bombs.
- Integrated real-time stdout/stderr stream parsing, syntax highlighting, and execution telemetry metrics.

AI Instagram Reel Operator – Desktop Automation Engine — **Python, Playwright, PySide6, Google Gemini, SQLite**

- Created a Windows automation system automating daily Instagram Reel creation, script generation, validation, and scheduled upload.
- Automated vertical 9:16 script generation with ChatGPT and video asset creation using Google Gemini via Playwright persistent sessions.
- Built media validation via ffprobe/ffmpeg trimmer, rotating file logging, and SQLite duplicate prevention tracking.

Grand Luxury Hotel Management – ERP & Booking Engine — **Next.js, PostgreSQL, Tailwind CSS, Google Sheets API**

- Designed digital guest registration, room reservation, and billing solution for boutique hospitality operations.
- Implemented temporal range exclusion constraints (PostgreSQL tsrange + GiST) mathematically preventing overbooking anomalies.
- Built automated guest invoice generation and bidirectional synchronization with Google Sheets for offline operational resilience.

PRACTICAL EXPERIENCE

Independent Software Developer | **Personal & Client Solutions** — **April 2025 – Present | Bijnor, UP, India**

- Designed and shipped 6 end-to-end applications spanning full-stack web development, native Android, and automation tooling.
- Architected relational database schemas in PostgreSQL with strict Row Level Security policies, foreign keys, and transaction atomicity.
- Implemented agentic software workflows utilizing modern AI developer tools with rigorous manual static analysis, testing, and security audits.

EDUCATION

Bachelor of Computer Applications (BCA) — **VKIT, Bijnor (Affiliated with GJUM, Moradabad)** — **2024 – Present (Currently Pursuing)**

- **Core Coursework:** Data Structures & Algorithms (C/C++), RDBMS & Database Design (SQL), Object-Oriented Programming (Java), Operating Systems, Computer Networks, Software Engineering Methodologies.